

# A physically consistent representation of recurring Jökulhlaup floods in a fully-coupled 2D subglacial hydrology model

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**ABSTRACT.** The design for the *Journal of Glaciology* has been implemented as a  $\text{\LaTeX 2\epsilon}$  class file and is derived from `article.cls`. We recommend that authors use this guide as a template. While writing we suggest you use the two-column `[twocolumn]` option to check that mathematical equations fit the measure. Submitted papers must, however, be presented using the one-column `[review]` option. If you have any problems using the class file, please contact Overleaf support at [www.overleaf.com/contact](http://www.overleaf.com/contact). The abstract should be less than 200 words and one paragraph long.

## 1 INTRODUCTION

1. Importance of ice-marginal lake drainage and potential hazard implications
2. Efforts to model
3. shortcomings and opportunities to improve

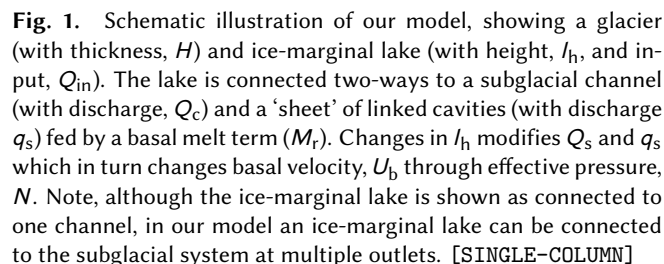
Ice-marginal lakes, glacier abutting water bodies dammed by ice or moraine, are increasing in size and frequency as a result of basin geometry changes driven by glacier thinning and retreat (Clague and others, 2012; Shugar and others, 2020; Mölg and others, 2021). Around Greenland alone, >3,000 ice-marginal lakes have been recorded (Carrivick and others, 2022), and their potential to repeatedly and catastrophically drain (known as

jökulhlaup or glacier outburst floods) pose major risks to downstream communities (Ng and Björnsson, 2003). In addition, jökulhlaup from ice-marginal lakes can impart significant influence on ice-dynamics (Carrivick and others, 2022; Dømggaard and others, 2024; Harpur and others, 2025; Livingstone and others, 2025), and Pleistocene jökulhlaup are considered large enough to have precipitated abrupt changes in global ocean circulation and climate (Teller, 1995; Clark and others, 2001; Mangerud and others, 2004; Süfke and others, 2022). Understanding the timings and triggers of jökulhlaup is imperative.

Numerous attempts to realise a numerical jökulhlaup theory have been made, all based upon mass conservation within a subglacial drainage system. Nye (1976) presented the first modern thermomechanical description, in which an ice-dammed lake drains via a single semi-circular subglacial

, the channel is enlarged by the heat dissipated by water flow leading to a positive feedback in which the rate of melting increases the rate at which the channel enlarges

We used the GlaDS model (Werder and others, 2013) as integrated into ISSM (Larour and others, 2012). GlaDS itself is a widely used (REF) 2D continuum model of subglacial hydrology. We refer interested readers to Appendix A.1 for an overview of key GlaDS equations, and to Werder and others



(2013) for a full description of the equations governing water flow. In summary, GlaDS operates on an anisotropic mesh generated using ISSM and includes a description of i) distributed flow through a series of linked cavities represented as a continuous sheet with variable thickness, and ii) channelised flow through R  thlisberger channels that are represented along element edges. Sheet elements exchange water with channels, and the cross-sectional area of these channels evolves through time due to the dissipation of potential energy, sensible heat exchange, and cavity closure rates due to viscous ice creep. Discharge in both the distributed ( $q_s$ ,  $\text{m}^2\text{s}^{-1}$ ) and channelised system ( $Q_c$ ,  $\text{m}^3\text{s}^{-1}$ ) is driven by gradients in the hydraulic potential,  $\nabla\phi$  ( $\text{Pa m}^{-1}$ ), with steeper hydraulic gradients driving enhanced water discharge. In GlaDS,  $Q_c$  is always non-zero along every internal edge, though only a few reach a discernible discharge. For ease of visualisation we follow previous work [REF] in declaring edges where  $Q_c \geq 0.5 \text{ m}^3\text{s}^{-1}$  to be ‘meaningful channels’ (e.g., Figure 2c–e). ISSM is a finite element higher-order ice-flow model which operates on anisotropic meshes and can be fully-coupled to additional modelling components (Larour and others, 2012). In addition to the ice-marginal lake modifications, which we describe below (Section 2.2), the primary difference between the original GlaDS (Werder and others, 2013) and the ISSM version used here is that rather than assuming water flow within the sheet to always be turbulent (cf. Werder and others, 2013), we make

use of the Hill and others (2024) transition parametrisation, whereby sheet flow switches between laminar and turbulent flow according to the local Reynolds number.

To capture the influence of ice-marginal lake drainage on ice dynamics, we include a two-way coupling between ice flow and subglacial hydrology. The GlaDS derived effective pressure ( $N = p_i - p_w$ , where  $p_i$  is ice pressure and  $p_w$  is water pressure, *Pa*) modifies basal velocity  $U_b$  ( $\text{m a}^{-1}$ ) by governing frictional resistance at the ice-bed interface. In turn, the magnitude of  $U_b$  modifies the rate at which cavities open and thereby controls the thickness of the sheet (Eq. A6–A7\*). Two-way coupling between ISSM and GlaDS is a relatively new development, and the influence of friction parameters remains the subject of ongoing research (e.g., Khan and others, 2024, ADD MORE REF), though model response is known to be strongly influenced by the choice of friction law (Choi and others, 2023). We view this work as necessarily explorative [Flesh this out more], and we do not undertake a full analysis of the modelling interactions between subglacial hydrology and ice dynamics, which is beyond the scope of this work. Rather, we seek to capture the broad influence ice-marginal lake drainage cycles have upon ice dynamics. Accordingly, we present results using two widely applied friction laws: i) the ‘Budd-style’ friction law [REF] implemented in ISSM in terms of basal stress with the form:

$$\tau_b = C_b^2 N^r U_b^s, \quad (1)$$

where  $\tau_b$  is the basal stress magnitude (UNITS),  $C_b$  is the friction coefficient,  $N$  is effective pressure,  $U_b$  is basal velocity magnitude, and both  $r$  and  $s$  are friction coefficients, or ii) a Schoof-style friction law [REF] with the form:

$$\tau_b = \frac{C_s^2 v_b^m}{\left(1 + \left(\frac{C_s^2}{C_{\max} N}\right)^{1/m} U_b\right)^m}, \quad (2)$$

where  $C_s$  is the Schoof friction coefficient,  $m$  is a friction law exponent, and  $C_{\max}$  is Iken’s bound.

## 2.2 Lake evolution

Following Kingslake and Ng (2013) lake depth,  $l_h$  (m, Fig. 1), evolves according to the continuity equation:

$$\frac{dl_h}{dt} = \frac{1}{A_L} [Q_{in} - Q_r(t-1)], \quad (3)$$

where  $A_L$  is the uniform lake area ( $\text{m}^2$ ),  $Q_{in}$  ( $\text{m}^3\text{s}^{-1}$ ) represents extraglacial lake input (e.g., rainfall, surface runoff) and  $Q_r$  ( $\text{m}^3\text{s}^{-1}$ ) represents inflow/outflow at the lake outlet(s) from the previous timestep. In moving from a 1D to 2D

representation of subglacial hydrology we include the contribution from the sheet, so that  $Q_r$  in our model is given by:

$$Q_r(t) = \sum_{i=1}^n (q_{s,i}(t) + Q_{c,i}(t)), \quad (4)$$

or the sum of the width integrated sheet discharge,  $q_s$ , and channel discharge,  $Q_c$  along  $n$  vertices connected to an ice-marginal lake. In GlaDS, a specific hydraulic potential is prescribed at outlets (Dirichlet boundary conditions), which is typically equivalent to zero water pressure (i.e., atmospheric pressure, Werder and others, 2013). As a boundary condition at lake outlets, hydrostatic lake pressure fixes the lake outlet hydraulic potential,  $\phi_{\text{outlet}}$  to be:

$$\phi_{\text{outlet}}(t) = \rho_w g z_b + \rho_w g l_h(t), \quad (5)$$

at all times, where  $g$  is gravitational acceleration ( $\text{m s}^{-1}$ ) and  $z_b$  is bed elevation (ma.s.l.). The evolving  $\phi_{\text{outlet}}$  handles the coupling between subglacial hydrology and ice-marginal lake by modifying the local  $\nabla\phi$  and therefore changing sheet/channel discharge (see Eq. A4 & A10).

## 2.3 Method of numerical solution

We solve for  $l_h$  using a Picard fixed-point iteration scheme comprising an outer and inner loop. In the outer loop, we update  $l_h$  given  $Q_r$  from the previous timestep (Eqn. 3) using a backward Euler solver and use this to set  $\phi_{\text{outlet}}$  (Eqn. 5) at the current timestep. In the inner loop, GlaDS solves for the domain-wide  $\phi$  until it satisfies a set convergence criteria, as in previous implementations of GlaDS (see: Werder and others, 2013; Hill and others, 2024). Following the inner loop converging, and given the global  $\phi$ , the model updates  $h_s$ ,  $N$ ,  $Q_c$ , and  $q_s$ , before summing  $Q_c$  and  $q_s$  at lake outlet nodes to give  $Q_r$  (Eqn. 4). The outer loop iterates until the norm of the current and previous  $l_h$  satisfies a given tolerance.

## 3 MODEL APPLICATION TO A SYNTHETIC DOMAIN

We test our model on a simple  $10 \times 3$  km rectangular domain (Figure 2a), discretised on a 2D triangular mesh with a characteristic edge length of 100 m ( $n_{\text{vertices}}=2486$ ). The bed elevation,  $h_b$  is characterised by a gentle slope ( $\sin = 0.0075$ ) dipping towards the terminus, and ice thickness is parabolic ( $H \propto \sqrt{x}$ ) increasing from  $H=50$  m at  $X=0$  km to  $H \sim 155$  m at  $X=10$  km. A  $5 \text{ km}^2$  ice-marginal lake was connected to a single vertex at the upper boundary of the domain where  $(X, Y) = (10, 1.5)$  km. To facilitate comparison

**Table 1.** Variables and Units<sup>1</sup>

| Variable | Units                      | Description                   |
|----------|----------------------------|-------------------------------|
| $\phi$   | Pa                         | Hydraulic potential           |
| $Q_c$    | $\text{m}^3 \text{s}^{-1}$ | Channel discharge             |
| $q_s$    | $\text{m}^2 \text{s}^{-1}$ | Sheet discharge               |
| $h_s$    | m                          | Sheet thickness <sup>a</sup>  |
| $l_h$    | m                          | Lake Height                   |
| $Q_r$    | $\text{m}^3 \text{s}^{-1}$ | Lake discharge                |
| $U_b$    | $\text{m a}^{-1}$          | Magnitude of basal velocity   |
| $u_b$    | $\text{m a}^{-1}$          | x-component of basal velocity |
| $v_b$    | $\text{m a}^{-1}$          | y-component of basal velocity |
| $S$      | $\text{m}^2$               | Channel cross-sectional area  |
| $t$      | yrs                        | Time coordinate               |
| $X$      | km                         | Horizontal coordinate         |
| $Y$      | km                         | Vertical coordinate           |
| $N$      | Pa                         | Effective pressure            |

<sup>a</sup>Note this includes the elastic sheet and cavity sheet thickness, see Eq. ??–A6

<sup>a</sup>The first section lists dependent variables, the second section lists coordinates, and the third section lists derived variables [FIX ORDER.]

with previous work, our domain is similar in scale and geometry to that in Kingslake and Ng (2013), which was chosen to be loosely analogous to the Merzbacher Lake and South Inylchek Glacier, Kyrgyzstan (Ng and Liu, 2009).

Model variables are given in Table 1 and our synthetic model parameters are given in Table 2. The parameter dependency of GlaDS has already been extensively explored elsewhere [REF]. To efficiently explore the parameter dependence of our ice-marginal lake drainage model we limited our testing to those previously highlighted as critical to subglacial drainage (e.g., channel/sheet conductivity terms,  $k_c/k_s$ , the basal bump height,  $h_b$ , the channel sheet width,  $l_c$ , and the cavity spacing,  $l_r$ ) as well as the terms which control the input of water into both the subglacial system (basal melt rate), and the ice-marginal lake (lake refill rate,  $Q_{in}$ ). As is common-practice, we varied one parameter at a time between a minimum and maximum bound (Table 2) informed both by previous work [REF] and the limits of numerical stability. As reference, an additional simulation was taken as the approximate midpoint of each of these parameters (Fig. 2). Each simulation was run for a total of 15 years and had a 30 minute timestep. We used a Budd-style friction law (Eq. 1) with the coefficient  $C_b = 100$  and exponents  $r = 1/4$  and  $s = 1/4$ . The ice dynamic model was initialised with  $U_b = 100, \text{m a}^{-1}$ , with Dirichlet boundary conditions ( $U_b = 0, \text{m a}^{-1}$ ) at the terminus and stress-free Neumann boundary conditions elsewhere. The hydrology model was initialised with  $\phi$  equivalent to 90% of ice overburden pressure, and an initial sheet thickness,  $h_s$  equal to half the bump

**Table 2.** [NOTE, this fits in two-column mode!] Input parameters in our synthetic domain. The parameters we varied are highlighted in bold, with the reference case value given first and the range over which they varied in parentheses.

| Description                               | Symbol       | Value(s)                  | Units                                     |
|-------------------------------------------|--------------|---------------------------|-------------------------------------------|
| Gravitational acceleration                | $g$          | 9.81                      | $\text{m s}^{-2}$                         |
| Latent heat                               | $L$          | $3.34 \times 10^5$        | $\text{J kg}^{-1}$                        |
| Ice density                               | $\rho_i$     | 917*                      | $\text{kg m}^{-3}$                        |
| Freshwater density                        | $\rho_w$     | 1000                      | $\text{kg m}^{-3}$                        |
| Pressure melt coefficient                 | $c_t$        | $7.5 \times 10^{-8}$      | $\text{K Pa}^{-1}$                        |
| Heat capacity of water                    | $c_w$        | FOLLOW UP                 | $\text{J kg}^{-1} \text{K}^{-1}$          |
| Glen's $n$                                | $n$          | 3                         |                                           |
| Ice flow constant at the bed <sup>a</sup> | $A_b$        | $6.8 \times 10^{-24}$     | $\text{Pa}^{-n} \text{s}^{-1}$            |
| Bed elevation                             | $z_b$        |                           | m                                         |
| Ice thickness                             | $H$          |                           | m                                         |
| Budd friction coefficient                 | $C_b$        | 100                       |                                           |
| First friction law exponent               | $r$          | 1/4                       |                                           |
| Second friction law exponent              | $s$          | 1/4                       |                                           |
| Channel conductivity                      | $k_c$        | <b>0.1 (0.25–0.01)</b>    | $\text{m}^3 \text{kg}^{-1} \text{s}^{-1}$ |
| Sheet conductivity <sup>b</sup>           | $k_s$        | <b>0.002 (0.05–0.005)</b> | $\text{Pa s}^{-1}$                        |
| Englacial void ratio                      | $e_v$        | $1 \times 10^{-4}$        |                                           |
| First sheet flow exponent                 | $\alpha_s$   | 5/4                       |                                           |
| Second sheet flow exponent                | $\beta_s$    | 3/2                       |                                           |
| First channel flow exponent               | $\alpha_c$   | 5/4                       |                                           |
| Second channel flow exponent              | $\beta_c$    | 3/2                       |                                           |
| Transition parameter                      | $\omega$     | 1/2000                    |                                           |
| Basal bump height                         | $h_b$        | <b>0.1 (0.2–0.05)</b>     | m                                         |
| Channel sheet width                       | $l_c$        | <b>20 (50–5)</b>          | m                                         |
| Cavity spacing                            | $l_r$        | <b>5 (20–1)</b>           | m                                         |
| Basal melt rate                           | $M_r$        | <b>0.25 (5–0.1)</b>       | $\text{m a}^{-1}$                         |
| Lake refill rate                          | $Q_{in}$     | <b>10 (20–5)</b>          | $\text{m}^3 \text{s}^{-1}$                |
| Elastic sheet depth scale                 | $h_c$        | 0                         | m                                         |
| Elastic sheet exponent                    | $\gamma$     | 1                         |                                           |
| Uplift regularisation rate                | $h_\epsilon$ | $1 \times 10^{-6}$        | $\text{m Pa}^{-1}$                        |
| Regularising pressure                     | $N_\epsilon$ | $1 \times 10^4$           | Pa                                        |

<sup>a</sup>The bed is assumed to be at pressure melting point, but the ice column is assumed equivalent to  $-10^\circ \text{C}$  or  $A = 4.9 \times 10^{-25} \text{ Pa}^{-n} \text{s}^{-1}$

<sup>b</sup>Note that following Hill and others (2024) the units for  $k_s$  differ from those given by Werder and others (2013)

height,  $h_b$ . We prescribe  $\phi$  terminus  $X = 0 \text{ km}$  to be  $\approx 1 \times 10^4$  as our GlaDS Dirichlet boundary condition, though arbitrary, this does not qualitatively affect our results below. We impose zero water inflow as our Neumann boundary condition at the other three boundaries. The ice-marginal lake had an initial height of 30 m ( $l_h$  at  $t=0$ ). The evolution of the reference case through time is shown in Fig. 2, and the sensitivity of lake height ( $l_h$ ), flood discharge ( $Q_r$ ), and basal velocity magnitude ( $U_b$ ) through time is shown in Figs. 3 & 4).

### 3.1 Stable flood cycles

In our reference case, the ice-marginal lake undergoes recurrent, stable, and asymmetric lake filling and draining cycles (Fig. 2a). Likewise, each lake filling and draining cycle

cle is associated with a recurring flood with stable maximum amplitude and duration (Fig. 2b). The lake only partially drains each flood cycle, and never reaches a height necessary to sustain flotation ( $N=0$ , Fig. 2f). For the initial 5 years of the simulation, as the model adjusts to initial conditions, peak flood discharge increases with each successive flood, and the lake reaches successively lower post-flood lowstands (Fig. 2a,b). Thereafter, the flood cycles reach a stable periodicity (Fig. 2f) and clockwise hysteresis with flood cycles occurring approximately every 1.5 years.

The evolution of the subglacial channel system through a single flood cycle is shown in Fig. 2c–e (an animated version is shown in SUPP MOVIE), and the width-averaged basal velocity response over the same flood cycle and into the following is shown in Fig. 2g. At the start of a flood cycle (Fig. 2c), as the lake approaches its highstand, the system is characterised by limited channelised drainage and  $\nabla\phi$  generally perpendicular to the ice surface slope. Close to the lake outlet ( $X > 9\text{km}$ ), contours of hydraulic potential appear concave forming a ‘ridge’ at the lake outlet. In GlaDS, sheet discharge,  $q_s$ , is directed perpendicular to these contours and such ridges indicate sheet flow out of the lake. Four short ( $<100\text{ m}$ ), low-discharge ( $\leq 1\text{ m}^3\text{s}^{-1}$ ) channels remain open following the previous flood cycle, three at the terminus and one connected to the lake outlet.

During the lake filling phase, increasing  $l_h$  drives  $\phi_{\text{outlet}}$  up (Eq. 5), resulting in a steeper  $\nabla\phi$  across the domain and promoting channelised drainage close to the lake outlet [SUPP MOVIE]. During the lake drainage phase, initiated once  $Q_r$  begins to rapidly exceed the lake refill rate,  $Q_{\text{in}}$  (when  $l_h \approx 75\text{ m}$  and  $t \approx 7.9\text{ yrs}$ ) an arborescent channel structure originating from the lake outlet extends across the full-width of the domain, perpendicular to the contours of  $\phi$ , and therefore offset to the direction of ice-surface slope. Simultaneously, channels extend upglacier from the terminus, and contours of  $\phi$  become increasingly convex. At the flood peak channel tributaries coalesce into the three primary terminus channels which evacuate water from the system. As the lake level drops sharply during a flood,  $\phi_{\text{outlet}}$  drops, reducing  $\nabla\phi$ , and in turn reducing  $Q_c$  below the level necessary to sustain channels across the domain, including at the lake outlet. Once  $Q_r < Q_{\text{in}}$  the flood terminates and the cycle begins anew. Each cycle takes  $\approx 1.5$  years with the exponentially rising and falling limb of the flood lasting  $\approx 6$  months.

Basal velocity,  $U_b$  changes significantly throughout a flood cycle and tracking the development of a significant channelised drainage system (e.g., Fig. 2d). Our synthetic glacier gradually accelerating during the flood initiation phase before channelised drainage is widely established,

then abruptly decelerating  $\approx$  two months before the flood peak (Fig. 2g) once channels are well established. Ice flow acceleration propagates from the lake outlet towards the terminus, with maximum velocity sustained within 2 km of the lake outlet.

### 3.2 Parameter sensitivity

Fig. 3 shows the time series from a total of 14 simulations across the maximum (red lines), minimum (blue lines) parameter values with the reference ‘mid-point’ simulation in grey. The majority of simulations yield stable and recurring asymmetric lake filling and drainage cycles (Fig. 3a[i]–f[i]) with corresponding floods in channel discharge at the lake outlet,  $Q_r$  (Fig. 3a[ii]–f[ii]). Amongst the parameters tested, the clearest influence is on the amplitude of lake height and its rate of change, both of which in turn influence the timing of flood initiation and peak flood discharge, with higher amplitude flood cycles leading to higher peak flood discharges.

Increasing channel conductivity,  $k_c$ , in effect making channels more sensitive to  $\nabla\phi$  (see Eq. A10), leads to lower lake highstands, higher peak discharge, earlier flood onsets and total lake drainage. At maximum  $k_c$  the model becomes physically unstable—the lake drains completely each flood cycle as peak discharge and flood duration increase until  $\sim 12$  years into the simulation when  $Q_r$  increases rapidly and continuously. Conversely, at minimum  $k_c$ , the capacity of channels to drain the lake is inhibited and flood onset is delayed. Because channels grow slowly, peak  $Q_r$  is also dampened leading to relatively long flood cycle durations ( $\sim 5$  years).

Increasing the sheet conductivity,  $k_s$  increases the capacity of the sheet to accommodate water and delays transmission to channels, this results in reduced lake highstands and delayed flood initiation as the onset of classical channel growth is delayed. Reducing  $k_s$  increases the relative efficiency of channels and leads to more vigorous lake-fill drain cycles, which shorter and more intense floods as a result. Changing  $h_b$  has a similar effect to reducing  $k_s$ , although at the maximum bump height, the sheet becomes so efficient that it can accommodate all lake discharge and within 1 year the lake reaches a stable equilibrium where  $Q_r = Q_{\text{in}}$ .

Changing the contributing width of the sheet to the channel ( $l_c$ ) or the representative horizontal spacing of cavities ( $l_r$ ) has similar influence on lake drainage by altering the relative efficiency of channelised and sheet drainage. Reducing  $l_r$  and increasing  $l_c$  reduces the amplitude of lake height and lowers peak discharge, and vice versa when increasing  $l_r$  and reducing  $l_c$ .

Finally, increasing input of water into the either the glacier via basal melt rate,  $M_r$  or lake via  $Q_{\text{in}}$  enhances the

amplitude of  $l_h$  oscillations and in turn peak flood discharge.

In all simulations, the maximum lake height remains below the maximum ice thickness and does not surpass the effective pressure  $N = 0$  necessary to initiate ice flotation. In the majority of simulations the lake lowstand does not fall below  $\sim 10$  m, only draining completely when  $k_c = 0.25 \text{ m}^3 \text{ kg}^{-1/2}$ , and when  $h_b = 0.05$  m.

The characteristics of the lake filling-drainage cycle across our parameter space also strongly influences the resulting basal velocity signal, and Figure 4 shows the width-averaged velocity during at least one complete flood cycle for  $k_c$ ,  $k_s$ , and  $h_b$ . Each model follows the reference case with velocity peaking before peak discharge is reached, and dropping rapidly thereafter along the falling limb of a flood. In general, shorter more intense flood cycles lead to a more rapid and higher magnitude velocity response, whereas more broad and less intense flood cycles yield a similar velocity response, maintaining peak velocity for a longer duration. However, the highest basal velocity ( $> 300 \text{ m a}^{-1}$ ) is associated with the low  $k_c$  case (Fig. 4), characterised by muted lake filling-drainage cycles but a relatively high lake-level (Fig. 3a[i]).

## 4 MODEL APPLICATION TO A REAL DOMAIN

We applied our model to Isunnguata Sermia, an outlet glacier in West Greenland draining a catchment of  $\approx 16000 \text{ km}^2$ . The central drainage axis of Isunnguata Sermia is characterised by a deep, flow-parallel trough with an anabranching structure in its upper reaches. A recently described  $\approx 3 \text{ km}^2$ ,  $\approx 100$  m deep ice-dammed lake (Dømsgaard and others, 2024; Livingstone and others, 2025) abuts the northern margin of Isunnguata Sermia's tongue. The lake drains subglacially with a periodicity of 1–3 years, undergoing at least 12 fill-drain cycles in the observational period between 1987–2024 (Livingstone and others, 2025). Two clear ice-contact boundaries are visible between Isunnguata Sermia and its ice-marginal lake (Figure 5), with the lake predominately draining subglacially along a contact at the western lake margin, and refilling by a proglacial river at the eastern lake margin (Livingstone and others, 2025).

We discretised our Isunnguata Sermia domain (Source of original catchment?) using an anisotropic mesh with an edge length of XX m close to terminus, dropping to XX km close to the upper margin of our domain at the approximate ELA (at  $H \approx 1600$  m). Bedmachine v5 is used for  $z_b$  and  $H$  (Morlighem and Al, 2022). As climate forcing, we used the 10 km resolution daily Modèle Atmosphérique Régional (MAR, v3.14) runoff and accumulation data (Fettweis and oth-

ers, 2017) between 2013–2024. An initialisation  $U_b$  was taken from the 2021 MEaSUREs annual mosaic (Joughin and others, 2015), from which we inverted for the basal friction coefficient using an L-curve scheme (following; Wolovick and others, 2023). Because we have an observational constraint on ice-surface velocity, we used a Schoof-style friction law (see Eq. 2, Section 2.1) which has been shown to more faithfully represent ice-dynamics with a GlaDS derived  $N$  than the Budd-style friction law used in our synthetic experiments (McArthur and others, 2023). Geothermal heat flux was taken from (Shapiro and Ritzwoller, 2004), basal melt rates were taken from Karlsson and others (2021) and surface ice temperature was taken from (Ettema and others, 2009).

Our GlaDS parameters are given in Tables 2 & A.1, with changes to the specific values intended to reflect the anticipated change in the scale of basal drainage elements relative to our synthetic domain (Hill and others, 2024). As boundary conditions, we set  $\phi$  equivalent to atmospheric pressure at the lowest elevation vertices within a cluster adjacent to a proglacial river at the glacier terminus (the cyan triangle in Fig. 5c), and set two lake-ice contact boundaries at the lowest elevation nodes along the western and eastern lake margin (the magenta stars in Fig. 5c).

Our experimental approach was split into two phases, the first of which was to obtain a basal friction coefficient  $C$ . We ran an initial inversion for  $C$  using an empirically-derived  $N$  taking into account bed topography and ice thickness, before running a simple GlaDS-only model run over 10 years, which did not include an ice-marginal lake, surface melt or two-way coupling to ice dynamics. This allowed us to derive a more realistic and spatially variable  $N$  from which a second inversion for basal friction could be obtained. Such an approach has previously been demonstrated to yield more faithful basal inversions in a coupled hydrology model (see; Ehrenfeucht and others, 2023; McArthur and others, 2023).

Our primary model phase, in which we ran our model for a total of 45 years with a 5 minute timestep, consisted of a 'steady state' phase, followed by a 'spin-up' into a full transient simulation. In the steady state phase, we ran our model for 30 years, until the 95<sup>th</sup> percentile difference in sheet thickness between successive timesteps fell below  $1 \times 10^{-6}$  m. During the steady state phase, we allowed the ice-marginal lake to freely evolve from an initial height of  $l_h = 30$  m. The steady state phase allowed the model to reach a condition which approximates wintertime hydrological conditions where there is limited channel activity and water pressures across the domain are close to the flotation fraction ( $p_w \approx 0.7-0.9p_i$ ). In the transient phase, taking the results of the steady-state run as initial conditions, we spun

**Table 3.** Parameters in our Greenland run, listing those which differ from Table 2

| Variable | Units | Description |
|----------|-------|-------------|
|----------|-------|-------------|

our model up over five years with a climate forcing that was gradually introduced by increasing the 2013 runoff by 20% each year. Following this spin-up, the full transient simulation proceeded from 2013 onwards using the unmodified MAR timeseries from 2013 until the end of 2024.

The full transient phase of our Greenland simulation is shown in Fig 5. The spin-up phase (dashed line in Fig 5e,f) is characterised by the development of a catchment wide sub-glacial drainage system together with a gradual increase in lake height during successively more intense melt summers (Fig 5d) before the first drainage in 2010. From 2013 onwards, once the model is subject to full-annual melt-forcing, the lake repeatedly fills and drains with a recurrence interval between floods of ~2 years. As in the synthetic case, lake fill and drainage cycles are characterised by a gradual lake filling phase and a rapid lake drainage phase, though the signal is notably more complex. Lake drainages typically occur at the end of a melt season. Lake levels then remain low during the subsequent winter, before filling in a stepwise fashion, with the lake filling more rapidly during summer melt seasons and remaining stable, or lowering slightly, during the intervening winter period. Lake drainage is triggered following the second melt season, and the cycle begins anew.

Unlike in the synthetic case, and likely reflecting variability in the meltwater forcing, the recurrence of floods is not stable, with flood characteristics variable year-on-year.

5 DISCUSSION

model reproduces 1) many observed characteristics of Jökulhlaup theory, 2) matches well with Felix and Jonny’s work, and 3) successfully reproduces many of the characteristics of IS flooding

However it is not perfect, initial conditions are a key uncertainty, as well as the other physically uncertain GlaDS parameters, noting we only carried out a partial sweep here, others were run but had limited influence on drainage characteristics . Something akin to Tim’s Gaussian GP emulation might offer a way to more rapidly test the parameter space and select an optimum starting condition.

Future work

6 CONCLUSIONS

7 AUTHOR CONTRIBUTION STATEMENT

The research was conceptualised by A. Hepburn, S. Buzzard, A. Sole, and S. Livingstone. A. Hepburn developed the model, carried out the experiments, and prepared the analysis. A. Hepburn wrote the manuscript, with input from S. Buzzard, A. Sole, and S. Livingstone. All authors reviewed and edited the manuscript, and fieldwork to deploy and maintain the sensors was carried out by everyone other than A. Hepburn. T. Irvine-Fynn is thanked for his assistance producing Figure 1. Fellowship funding was acquired by A.Hepburn and project funding by S. Livingstone, R. Storrar, S. Buzzard, A. Sole, [FINISH].

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9 PROPER NOUNS WILL BE CORRECTED IN REFERENCES PRE-SUBMISSION...

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## A ADDITIONAL MODEL DESCRIPTION

### A.1 The Glacier Drainage System model

As described fully in Werder and others (2013), the hydraulic potential at the bed,  $\phi$ , is defined at the bed by:

$$\phi = \phi_m + p_w, \quad (\text{A1})$$

with water pressure,  $p_w$ , and elevation potential

$$\phi_m = \rho_w g z_b, \quad (\text{A2})$$

where  $\rho_w$  is water density,  $g$  is gravitational acceleration and  $z_b$  is bed elevation. In turn, effective pressure  $N$  is given by:

$$N = p_i - p_w = \phi_0 - \phi, \quad (\text{A3})$$

where  $p_i$  is ice overburden pressure ( $p_i = \rho_i g H$ ) and  $\phi_0$  is the overburden potential ( $\phi_0 = \phi_m + p_i$ ).

In the original GlaDS (Werder and others, 2013), sheet discharge,  $q_s$  is given by:

$$q_s = -k_s h_s^\alpha |\nabla \phi|^\beta \nabla \phi, \quad (\text{A4})$$

where  $k_s$  is the sheet conductivity;  $h_s$  is the sheet thickness; and  $\alpha$  and  $\beta$  are flow exponents. Following the Darcy-Weisbach law, when flow is fully turbulent the flow exponents  $\alpha = 5/4$  and  $\beta = 3/2$ . Here we use the Hill and others (2024) laminar-turbulent transition model, and with  $\alpha_s = 5/4$  as here, Eq A4 becomes:

$$q_s = -\frac{\nu}{2\omega} \left( \frac{h_b}{h_s} \right)^{1/2} \cdot \left( -1 + \sqrt{1 + 4 \frac{\omega}{\nu} \left( \frac{h_s}{h_b} \right)^{1/2} k_s h^3 |\nabla \phi|} \right) \frac{\nabla \phi}{|\nabla \phi|} \quad (\text{A5})$$

where  $\omega$  is a laminar–turbulent transition parameter (corresponding to the Reynolds number at which water becomes fully-turbulent),  $h_b$  is the bed bump height, and  $\nu$  is the kinematic viscosity of water at 0°C.

The cavity sheet thickness evolves through time given by:

$$\frac{\partial h_{\text{cav}}}{\partial t} = w - r, \quad (\text{A6})$$

for functions  $w$  and  $r$ , which describe the cavity opening and closing rate, respectively (Kamb, 1987; Walder and Fowler, 1994). Cavities open at a rate given by basal velocity  $U_b$  over basal bumps with height,  $h_b$ :

$$w(h) = \begin{cases} U_b(h_r - h_s)/l_r & \text{if } h < h_r \\ 0 & \text{otherwise} \end{cases} \quad (\text{A7})$$

where  $l_r$  is the horizontal cavity spacing. Viscous ice deformation leads to cavity closure, related to the effective pressure,  $N$  by:

$$r(h, N) = A h |N|^{n-1} N, \quad (\text{A8})$$

where  $A$  is the rheological constant of ice, multiplied by a first order geometrical factor ( $h$ , but double check it is sheet thickness), and  $n$  is Glen's flow law exponent.

Sheet elements exchange water with channels, the cross-sectional of which,  $S$ , evolve through time due to the dissipation of potential energy,  $\Pi$ , sensible heat exchange,  $\Xi$ , and cavity closure rates due to viscous ice creep  $v_c$

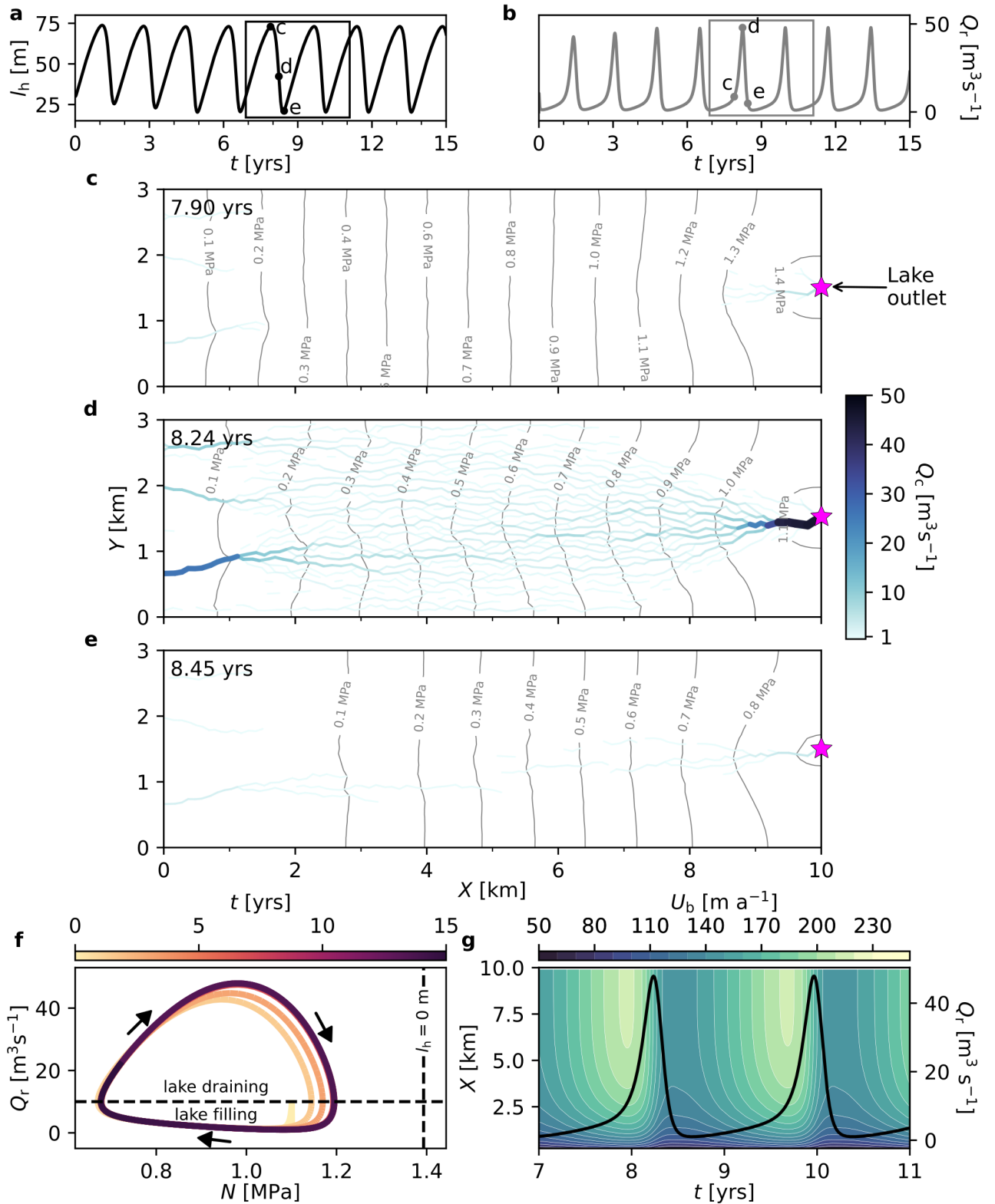
$$\frac{\partial S}{\partial t} = \frac{\Xi - \Pi}{\rho_i L} - v_c, \quad (\text{A9})$$

where  $\rho_i$  is the ice density and  $L$  is the latent heat of fusion.

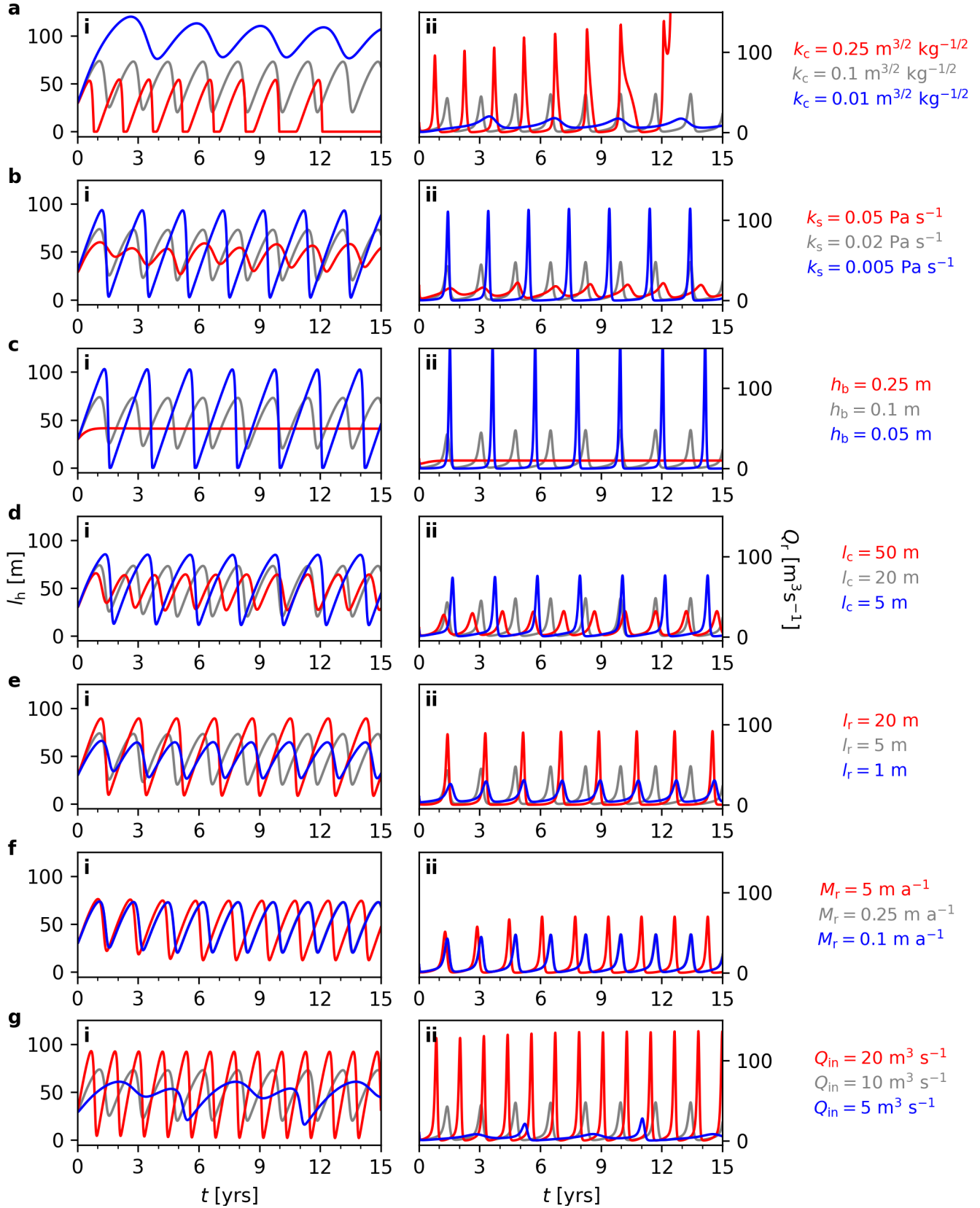
Finally, channel discharge,  $Q_c$  is also related to the hydraulic potential gradient by:

$$Q_c = -k_c S^{\alpha_c} \left| \frac{\partial \phi}{\partial s} \right|^{\beta_c - 2} \frac{\partial \phi}{\partial s}, \quad (\text{A10})$$

where  $k_c$  is the channel conductivity;  $s$  is the horizontal coordinate along a channel; and  $\alpha_c/\beta_c$  are channel flow exponents.



**Fig. 2.** Evolution of the 'reference' model configuration in our synthetic modelling domain. **a)** Lake height,  $l_h$  through time. Black points indicate  $l_h$  in panels c–e. **b)** Sum discharge at the lake outlet,  $Q_r$  through time. Grey points indicate  $Q_r$  in panels c–e the grey box indicates the time span of panel g. **c–e)** Channel discharge,  $Q_c$  at  $t = 7.10$  years (c), 8.38 years (d) and 9.00 years (e). In each panel, effective pressure,  $N$  is shown as grey contours and the lake outlet is marked by a magenta star. **f)** Phase-plane diagram showing the evolution of  $Q_r$  and  $N$  through time at the ice-marginal lake outlet. Arrows and shading indicate the direction of time. The horizontal dashed line denotes the lake refill rate,  $Q_{in} = 10 \text{ m}^3 \text{ s}^{-1}$ . The vertical dashed line indicates  $N$  when  $l_h = 0$  m. **g)** Filled contour plot showing the across-flow averaged velocity through time. White contours represent velocity (every  $10 \text{ m a}^{-1}$ ) and the black line shows  $Q_r$  between years 7–11.



**Fig. 3. NOT FINAL VERSION!** Lake height,  $l_h$  (left panels, i) and sum discharge at the lake outlet,  $Q_r$  (right panels, ii) across a range of key model parameters. **a)** Channel conductivity,  $k_c$ . **b)** Sheet conductivity,  $k_s$ . **c)** Basal bump height,  $h_b$ . **d)** Channel sheet width,  $l_c$ . **e)** Cavity spacing,  $l_r$ . **f)** Basal melt rate,  $M_r$ . **g)** Lake refill rate,  $Q_{in}$ . In all panels, the default parameter run is shown in grey.

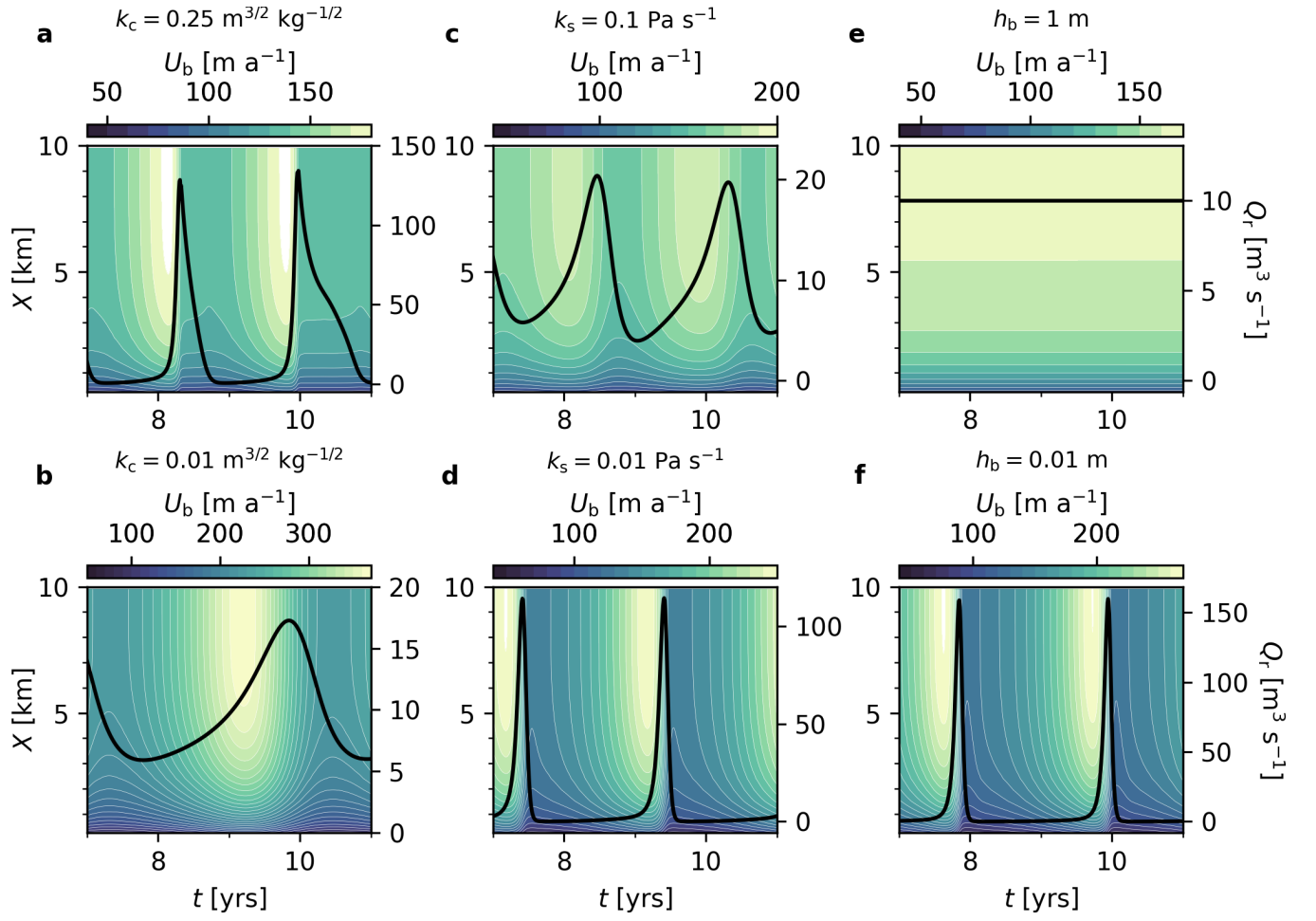
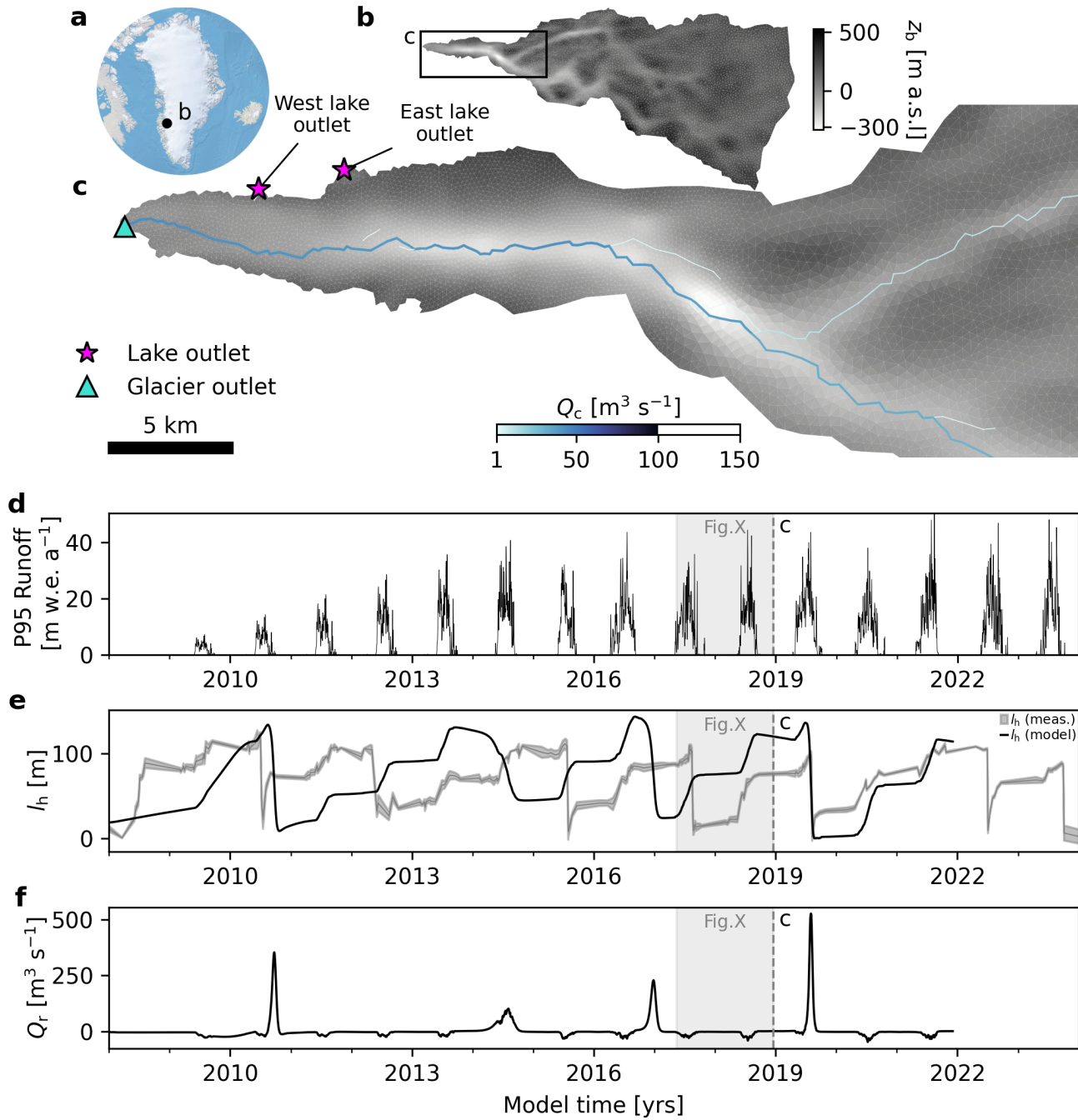
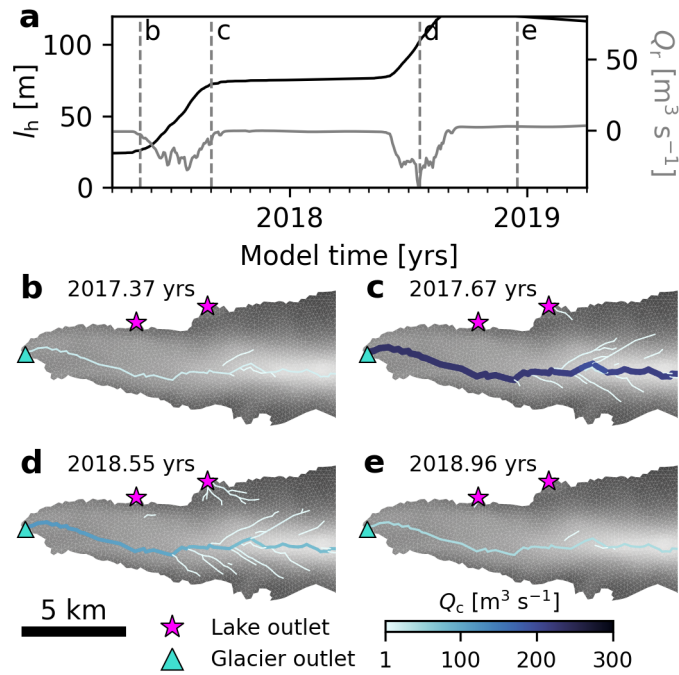


Fig. 4. NOT FINAL VERSION!



**Fig. 5.** Ice-marginal lake drainage in the Isunnguata Sermia glacier catchment. **a)** the location of our model domain (panel b) in southwest Greenland. **b)** our modelling domain, coloured by the bed elevation (Morlighem and AI, 2022), the extent of panel c is outlined. **c)** channel discharge,  $Q_c$  at  $t = XXX$  yrs. The location of lake outlets and the glacier outlet are marked by pink stars and a cyan triangle respectively. The primary drainage axis is located within the along-flow orientated subglacial trough. **d)** 95<sup>th</sup> percentile runoff during the transient modelling phase. Runoff increases by 20% each year during the spin-up phase (2008–2013). **e)** measured lake height,  $l_h$  (shown in grey with shaded error bars), and modelled  $l_h$  (black line) during the transient phase. Measured  $l_h$  and uncertainty values taken from Livingstone and others (2025). **f)** sum outlet discharge,  $Q_r$  at the lake outlets during the transient phase. Values lower than zero indicate net discharge into the lake, and vice versa when  $Q_r > 0$ . The dashed vertical line in panels d–f shows the time represented by panel c, and the shaded grey box indicates the time span shown in Fig. 6.



**Fig. 6. NOT FINAL VERSION!** Channel development during a complete lake filling–drainage cycle. **a)** lake height,  $l_h$  and flood outlet discharge,  $Q_r$  through time. **b–e)** snapshots of channel discharge,  $Q_c$  through time, with the location of lake outlets and the glacier outlet marked by pink stars and a cyan triangle respectively. Dashed lines in panel a indicate the snapshot times in panels b–e. [SINGLE-COLUMN]